

# Alice - the semiotic state machine LLM

**Gary Nan Tie, Sep 21, 2026**

## **Abstract**

We introduce a refinement of a state machine LLM that combines a recurrent implicit deep learning model transition with an attention-based output over learned key-value fixed points, optimized for syntactic prediction and semantic coherence within a word embedding space. Beyond merely predicting text the novel architecture explicitly constructs semantic state trajectories, tests entailment/contradiction, performs abduction, and enables a semiotic channel between AI agents. This machine epistemology allows us to ask: what follows, what is excluded and what could explain it. Logical reasoning via the semiotic hexagon and genetic algorithm selection make possible recursive swarm improvement towards thinking machines.

## Synopsis

We define (in Appendix) a novel state machine  $(S, O)$

via an implicitRNN: For  $i \in [m]$ , tokens  $t_i \in \mathbb{R}^p$ .

Initial condition:  $h_0 := 0 \in \mathbb{R}^q$

Input:  $u_i := [t_i \ h_{i-1}]'$

First equilibrium equation:

$$k_i = \phi(A k_i + B u_i)$$

Second equilibrium equation:

$$v_i = \psi(C v_i + D k_i)$$

Prediction equation:

$$h_i = S(h_{i-1}, t_i) := E v_i + F k_i$$

(where  $A := \Gamma - \text{diag}(|\Gamma| 1) + \gamma \text{Id}$ ,  $0 < \gamma < 1$ , ensures  $A$  well-posed for nonlinear activation  $\phi$ , that is, the fixed point equation has a solution and it is unique, similarly  $C$  is well-posed for activation  $\psi$ ).

Attention layer:

Input  $h_i$ .

For  $i \in [m]$ , define probabilities  $\alpha_{i,j} = \text{softmax over } j \in [i] \text{ score}(h_i, k_j)$  where score is a similarity function.

Let  $h^i = \text{Sum over } j \in [i] \alpha_{i,j} v_j$  and

define update  $h'_i = \text{FFN}_\theta(h^i)$ .

Output  $h'_i$ .

Stack  $L$  attention layers with parameters  $\Theta = \{\theta_1, \dots, \theta_L\}$ , composing updates  $\text{FFN}_\theta$  and now let  $h'_i$  denote the final  $L$ -th update. After  $L$  attention layers, for  $i \in [m]$  define probabilities  $\beta_{i,j} = \text{softmax over } j \in [i] \text{ score}(h'_i, v_j)$  and let  $j^*_i = \arg \max \text{ over } j \in [i] \beta_{i,j}$ . Define token continuation  $o_{i+1} = O(h_i) := t_{(j^*_i)}$ . This is how attention layers construct output  $O$  corresponding to transition  $S$ .

Define average Renyi  $\alpha$ -entropy of  $\{t_i\}$  by:

$$E(\{t_i\}) = (1/m) \text{Sum over } i \in [m] H_i \text{ where}$$

$$H_i := H_\alpha(\{\beta_{i,j}\}) \text{ and } H_\alpha \text{ is Renyi } \alpha\text{-entropy.}$$

State machine  $(S,O)$  training problem:

Let  $\lambda > 0$ , learn matrices  $A, B, C, D, E, F, G, H$ , fixed points  $k_j, v_j$  and feedforward layer parameters  $\Theta$  to minimize:

$$\text{Sum over } i \in [m] \{1 - \text{Cos}(v_i, \text{Emb}(t_i))\} +$$

$$\text{Perplexity}(o_2, \dots, o_{m+1}) + \lambda E(\{t_i\})$$

where Emb is a word embedding, like word2vec.

Comments:

Fixed points  $(k_j, v_j)$  form a nonlinear key-value pairing.

State  $h_i$  is a contextual representation of token  $t_i$ .

Agent states can combine [6] to produce offspring via state combination operators like convolution, Kronecker product, Hadamard product, and rank-one operators, enabling genetic algorithm selection and evolution.

Value  $v_i$  simulates word embedding  $\text{Emb}(t_i)$ .

The objective is to minimize the perplexity of attention layer outputs  $o_i$  regularized by Renyi  $\alpha$ -entropy.

The training problem can be learned in stages, first the word embedding then attention.

The training problem can be parallelized [2].

State machine  $(S, O)$  is a LLM that combines a recurrent implicit deep learning model transition with an attention-based output over learned key-value fixed points, optimized for syntactic prediction and semantic coherence within a word embedding space. It is a refined variation (cf  $\beta$ ) on [1], [2] and [5].

Entailment [3]:

Suppose  $(S, O)$  processes two token string sentences, premise  $P = (p_1, \dots, p_r)$  and hypothesis  $H = (h_1, \dots, h_s)$  as follows:

$$h_{P,j} = S(h_{P,j-1}, p_j), j \in [r] \text{ and}$$

$$h_{Q,j} = S(h_{Q,j-1}, q_j), j \in [s],$$

to obtain  $h_P := h_{P,r}$  and  $h_Q := h_{Q,s}$ .

Let  $h'$  denote the final update in a stack of attention layers in output  $O(h)$  of  $(S, O)$ . Define  $\beta_j(h) = \text{softmax over } j \in [m], \text{score}(h', v_j)$  and let  $\beta(h) = (\beta_1(h), \dots, \beta_m(h))$  denote the distribution of  $h'$  over  $\{v_j\}$ .

Fix  $0 < \tau < 1$  and integer  $k$ .

Define  $F(h) := \{k_j \mid \beta_j(h) > \tau\}$ .

Define **entailment**  $P \Rightarrow Q$  iff

$F(h_Q)$  is a subset of  $F(h_P)$

Define **contradiction**  $P \Rightarrow \text{not } Q$  iff

$$|F(h_P) \cap F(h_Q)| < k$$

On training triples  $(P, Q, y)$ , where

$y \in \{\text{entail, neutral, contradiction}\}$

learn Entailment defined by:

Entailment( $h_P, h_Q$ ) :=  
 entail, if  $P \Rightarrow Q$ ,  
 contradiction, if  $P \Rightarrow \text{not } Q$ ,  
 neutral otherwise.

to minimize, Embedding loss, Perplexity loss regularized by average Renyi  $\alpha$ -entropy, and Entailment loss.  
 Thereby, (S, O) learns textual entailment.

Abductive inference [3]:

Given a hypothesis Q, we wish to find a premise P such that  $P \Rightarrow Q$ . Token string Q may be auto-regressively continued by (S, O) to generate Q, P1, P2, ... , Pn, where  $n > 1$ , and Pj denotes a continuation sentence. Choose  $0 < \tau < 1$ , for each  $j \in [n]$ , check if  $P_j \Rightarrow Q$ .

Suppose  $\Omega = \{ P_j \mid P_j \Rightarrow Q \}$  is non-empty.

Let  $m := \min \text{ over } P \in \Omega, |F(h_P)|$  and  $P_j^*$  be the first Pj In  $\Omega$  with  $|F(h_{P_j})| = m$ .

Abduction: Given a hypothesis Q, we want to determine a premise P that entails Q. If  $\Omega$  is non-empty, choose premise  $P_j^*$  as above that entails Q.  $P_j^* \Rightarrow Q$  is a parsimonious selection of a premise to explain Q.

So LLM (S,O) has textual entailment and abductive reasoning.

Semiotics [4]:

Suppose random variable X has distribution  $p_X$ , denoted  $X \sim p_X$ .

Let  $H_\alpha$  denote Renyi  $\alpha$ -entropy.

For  $(X, Y) \sim p_{X,Y} = p_X p_{Y|X}$ , Arimoto mutual information  $I_{A,\alpha}(X; Y) := H_\alpha(X) - H_{A,\alpha}(X|Y)$  where  $H_{A,\alpha}$  is Arimoto conditional  $\alpha$ -entropy.

To quantify the uncertainty of text, define semiotic breadth  $S(Q) := H_\alpha(h_Q)$  where  $h_Q \sim \beta(h_Q)$ .

We say P is an interpretation of Q if  $P \models Q$

and define semiotic decipherability

$D(P) := I_{A,\alpha}(h_Q; h_P)$

to quantify the reduction in uncertainty about an interpretation that results from observing a message.

To profile semiotic channel capacity  $C_{A,\alpha} := \max \text{ over } h_Q, I_{A,\alpha}(h_Q; h_P) \text{ map } S \text{ against } D,$

creating a LLM semiotic cloud signature.

Recall the theory of opposites in semiotics, in particular the Greimas square and its extension to the logical hexagon. The square captures contrary, contradictory, and subaltern relations, while the hexagon adds existential and mixed-state positions. As LLM (S, O) has contradiction, namely  $P \models \text{not-}Q$ , we can also distinguish meanings from their opposites. Hence AI agents using this LLM channel have three complementary operations: Entailment, what follows?; Contradiction, what is excluded?; Abduction, what could explain it? Moreover, the semiotic logical hexagon enables logical reasoning.

In summary, entailment, contradiction, abduction, logical reasoning, semiotic communication, interpretability, and emergent agent evolution through natural selection are all possible within this computational epistemic system. This is where AI safety concerns are heightened, because the system's search space is no longer simply "what kind of response may best be generated?" but potentially "what kind of state machine may evolve?" [6], such as recursive swarm improvement towards thinking machines. The most important caveat is that none of these ramifications follows automatically from the architecture. They are research hypotheses enabled by the architecture. In particular, one would need to conduct careful experiments to parse the interactions between these constructs and ensuing inter-agent behavior.

## Semiotic LLM ALICE

— Abductive Learning and Implicit Contextual Entailment.

## Appendix - Semiotic State Machine LLM

Corpus  $\{t_1, \dots, t_m\}$ , token  $t_i \in \mathbb{R}^p$

Word embedding  $\text{Emb}: \mathbb{R}^p \rightarrow \mathbb{R}^{p'}$

Transition  $\mathcal{S}$  defined by an implicit RNN:

Initial state  $h_0 = 0 \in \mathbb{R}^q$

Input  $u_i = \begin{bmatrix} t_i \\ h_{i-1} \end{bmatrix} \in \mathbb{R}^r$ ,  $r = p + q$ ,  $i \in [m]$

$k_i = \varphi \left( \underset{q \times q}{A} k_i + \underset{q \times r}{B} u_i \right) \in \mathbb{R}^q$ ,  $A$  well-posed for  $\varphi$

$v_i = \gamma \left( \underset{p' \times p'}{C} v_i + \underset{p' \times q}{D} k_i \right) \in \mathbb{R}^{p'}$ ,  $C$  well-posed for  $\gamma$

$h_i = \mathcal{S}(h_{i-1}, t_i) \triangleq \underset{q \times p'}{E} v_i + \underset{q \times q}{F} k_i \in \mathbb{R}^q$

—

Let  $\Theta = \{(W_1, b_1), (W_2, b_2)\}$  be parameters for a

feedforward network defined by

$\text{FFN}_{\Theta}(h_i) \triangleq \text{ReLU}(h_i W_1 + b_1) W_2 + b_2$

Let  $\text{Cos}(a, b) \triangleq \frac{\langle a, b \rangle}{\|a\| \|b\|}$

Output  $\oplus$  defined by attention layers:

Attention layer:

Input  $h_i \in \mathbb{R}^f$

For  $i \in [m]$ ,  $\alpha_{ij} \triangleq \text{softmax}_{j \in [i]} \text{score}(h_i, k_j)$

where  $\text{score}$  is a similarity function.

Let  $\hat{h}_i \triangleq \sum_{j=1}^i \alpha_{ij} v_j$  and update  $h'_i \triangleq \text{FFN}_{\theta}(\hat{h}_i)$   $[\dim h'_i = \dim k_j]$

Output  $h'_i \in \mathbb{R}^g$

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Stack  $L$  attention layers with parameters

$\Theta = \{\theta_1, \theta_2, \dots, \theta_L\}$ , composing updates  $\text{FFN}_{\theta_L}$ ,  $L \in [L]$ .

Now let  $h'_i$  denote the final  $L$ -th update.

For  $i \in [m]$ ,  $\beta_{ij} \triangleq \text{softmax}_{j \in [i]} \text{score}(h'_i, v_j)$

and let  $j_i^{\beta} = \arg \max_{j \in [i]} \beta_{ij}$ .

Define token continuation  $o_{i+1} = \oplus(h_i) \triangleq t_{j_i^{\beta}} \in \mathbb{R}^p$

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Define average Renyi  $\alpha$ -entropy of  $\{t_i\}_{i=1}^m$  to be:

$$E(\{t_i\}_{i=1}^m) \triangleq \frac{1}{m} \sum_{i=1}^m H_{\alpha}^i$$

where  $H_{\alpha}^i \triangleq H_{\alpha}(\{\beta_{ij}\}_{j=1}^i)$  and  $H_{\alpha}$  is Renyi  $\alpha$ -entropy.  
→

State machine  $(\mathcal{S}, \mathcal{O})$  training problem:

Let  $\lambda > 0$ , learn matrices  $A, B, C, D, E, F$ ,

fixed points  $k_j, v_j$  and feedforward layer parameters  $\textcircled{H}$

to minimize:  $\sum_{i=1}^m (1 - \cos(v_i, \text{Emb}(t_i)))$

+  $\text{Perplexity}(o_2, \dots, o_{m+1}) + \lambda E(\{t_i\}_{i=1}^m)$

where  $\text{Emb}$  is a word embedding, like word2vec.  
→

LLM Alice  $(\mathcal{S}, \mathcal{O})$  has textual entailment  $\models$

and abductive inference [3], as well as semiotic

channel capacity [4].

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